NEXT ENGINEER · _SHEERAT_

Universal Human Values and Professional Ethics

BVE301

One-day revision notes: definitions, solved examples, code and important questions

Program: B.Tech | **Branches:** ALL | **Semester:** 3 | **Credits:** 3 | **L-T-P:** 2-1-0

Scheme: AKTU B.Tech 2nd year, 2023-24 onwards (also offered as BVE401 in semester 4)

UNIT 1	UNIT 2	UNIT 3	UNIT 4	UNIT 5
Introduction to Value Education	Harmony in the Human Being	Harmony in the Family and Society	Harmony in Nature and Existence	Holistic Understanding and Professional Ethics

Original revision notes written for the AKTU syllabus, which follows the framework of the prescribed textbook by Gaur, Asthana and Bagaria. Every unit has:

1. **2-mark definitions**: short answers you can write word for word in Section A.
2. **Core concepts**: the ideas, lists and diagrams that make up long answers.
3. **Examples from life**: short illustrations examiners like to see in answers.
4. **Key points to remember**: compact lists.
5. **Important questions**: the long questions to prepare, with what your answer must contain.

How to use these notes the day before the paper

- **Morning (2 hours):** Units 1 and 2. Learn the basic human aspirations, the three basic requirements and the Self–Body distinction. Draw the "I and Body" table.
- **Afternoon (2 hours):** Units 3 and 4. Learn the nine values in relationships, the four comprehensive human goals and the four orders of nature.
- **Evening (1 to 2 hours):** Unit 5 (ethical conduct, holistic technologies, transition strategy), then the "Last-hour checklist".
- **In the exam:** use the course's own terms with their meanings (Sukh–Suvidha, Sanyam–Swasthya, Vishwas, Samman, Sah-astitva...). Add one everyday example for each point, and a diagram where possible. Answers that only give opinions without the framework score less.

AKTU paper pattern (theory, 70 marks, 3 hours): Section A has 7 short questions of 2 marks, all compulsory (14). Section B has 5 questions of 7 marks, attempt any 3 (21). Section C has 5 questions (one per unit), each with two parts, attempt one part of each, 7 marks each (35). Every unit gets its own Section C question, so do not skip any unit.

UNIT 1 Introduction to Value Education

✍️ 2-mark definitions

- **Value:** the role or participation of a unit in the larger order it belongs to; for a human being, values are what make living fulfilling for oneself and others (e.g. trust, respect, care).
- **Value education:** education that helps us understand what is truly valuable for human happiness, and live accordingly. It complements skill education.
- **Skills vs values:** values tell us *what* to do (the goal); skills tell us *how* to do it. Both are needed: values without skills are ineffective, skills without values can be harmful.
- **Self-exploration:** a process of looking within to check whether our beliefs and desires are naturally acceptable to us, by verifying them in our own living rather than accepting them on authority.
- **Natural acceptance:** what we accept as right deep within, independent of time, place and person (e.g. we naturally accept respect, not disrespect).
- **Experiential validation:** checking a proposal by living it and seeing whether it leads to mutual fulfilment in behaviour and mutual prosperity in work.
- **Basic human aspiration: continuous happiness and prosperity**.
- **Happiness:** a state of being in harmony (*synergy*); being in a state we naturally accept.
- **Prosperity:** the feeling of having or producing more than the required physical facilities.

💡 Core concepts

Need for value education: to understand our aspirations correctly, to complement technology and skills with right goals, to evaluate our beliefs, and to move from conflict and confusion to harmony at every level.

Guidelines for value education: it should be **universal** (for all humans, all times and places), **rational** (based on reasoning, not blind belief), **natural and verifiable** (acceptable to the self and verifiable by experience), **all-encompassing** (covering all dimensions of life) and **leading to harmony**.

Content of value education: understanding harmony at all four levels: **individual, family, society and nature/existence**.

Process of self-exploration

1. Start with a **proposal** (not an assumption).
2. Check it on the basis of **natural acceptance**.
3. Verify it by **experiential validation** in living: behaviour with humans (leading to mutual happiness) and work with the rest of nature (leading to mutual prosperity).
4. If it passes, it becomes our own understanding.

Basic requirements for fulfilling human aspirations (with correct priority)

1. **Right understanding** (comes first)
2. **Relationship** with other humans

3. **Physical facilities** (needed, but last in priority)

Present scenario (critical appraisal): today most people give priority only to physical facilities, assuming that more wealth means more happiness. As a result:

- People who **lack** physical facilities feel deprived and unhappy.
- People who **have plenty** of physical facilities often still feel deprived and unhappy, and may try to exploit or dominate others to get even more.
- Living focused only on physical facilities is **animal consciousness**. **Human consciousness** means living with right understanding, relationship and physical facilities in the right priority. Only then can a person have right understanding, feel prosperous, and be happy in themselves and in relationships.

Methods to fulfil aspirations: understand and live in harmony at four levels: (1) in the self, (2) in the family, (3) in society, (4) in nature/existence.

Examples from life

- A student studies only for a high-paying job (physical facilities) but neglects relationships and understanding; even after getting it, they feel stressed and lonely. This shows that physical facilities alone do not give continuous happiness.
- When someone insults us, we feel unhappy even if we are rich; when someone trusts us, we feel happy even with little money. This shows that relationship matters more than wealth for happiness.

Key points to remember

```
Basic aspiration: continuous happiness and prosperity
Requirements (priority): right understanding → relationship → physical facilities
Self-exploration = proposal → natural acceptance → experiential validation
Value education guidelines: universal, rational, natural & verifiable, all-encompassing, leading to harmony
Harmony at 4 levels: self, family, society, nature
```

⭐ Important questions

1. What is value education? Explain its need and the basic guidelines for it.
2. What is self-exploration? Explain its content and process (natural acceptance and experiential validation). *Draw the process flow.*
3. What are the basic human aspirations? Explain the three basic requirements and their correct priority.
4. Critically appraise the current understanding of happiness and prosperity (animal consciousness vs human consciousness).
5. Differentiate between values and skills with examples.

UNIT 2 Understanding Harmony in the Human Being

✍️ 2-mark definitions

- **Human being = co-existence of "I" and Body:** "I" (the self, sentient, conscious, the one that knows, decides and experiences) and the Body (material, physico-chemical).
- **Sukh:** happiness, the need of the self ("I"); continuous in time, fulfilled by right understanding and right feelings.
- **Suvidha:** physical facilities, the need of the Body; temporary and limited in quantity.
- **Body as an instrument of "I":** "I" is the **seer (drashta), doer (karta) and enjoyer (bhokta)**; the Body only carries out the decisions of "I".
- **Activities of "I":** desire (imaging), thought (analysing), expectation (selecting/tasting); also realization and understanding.
- **Sanyam:** self-regulation: the feeling of responsibility of "I" for nurturing, protecting and rightly using the Body.
- **Swasthya:** health: the Body acting according to "I", with its parts in harmony.
- **Prosperity:** the feeling of having more than enough physical facilities; it needs right identification of needs and production of more than that.

💡 Core concepts

"I" vs Body (draw this table)

	"I" (Self)	Body
Nature	Sentient (conscious)	Material (physico-chemical)
Needs	Happiness (sukh), e.g. respect, trust	Physical facilities (suvidha), e.g. food, clothes
Needs in time	Continuous	Temporary
Needs in quantity	Qualitative (cannot be measured)	Quantitative (limited)
Fulfilled by	Right understanding and right feelings	Physical facilities
Activities	Knowing, assuming, recognizing, fulfilling	Recognizing, fulfilling

Confusion between needs of "I" and Body: we try to get happiness (a need of "I") through physical facilities (a need of the Body). Since physical needs are limited and temporary, this never gives continuous happiness; it leads to endless wanting, consumerism and stress.

Harmony in "I": when our desires, thoughts and expectations are guided by right understanding, they align with natural acceptance, and we feel at peace. Contradictions between them cause unhappiness.

Harmony of "I" with the Body: Sanyam and Swasthya

- Sanyam (in "I") leads to Swasthya (in the Body).
- **Programmes for Sanyam and Swasthya:** proper nutrition (*ahar*), regulated daily routine (*vihar*), physical work and exercise, *asana* and *pranayama*, treatment of illness (*aushadhi*), correct posture

and use of the body, avoiding misuse (intoxicants, overeating).

Correct appraisal of physical needs: physical needs are **limited and definite**. Prosperity means identifying these needs correctly and producing more than needed; then we feel prosperous and can share. Without this, even great wealth feels insufficient.

Examples from life

- Eating tasty food (suvidha) gives pleasure for a short time, but if a friend shares it with respect (sukh), the happiness lasts.
- A person who exercises and eats on time (sanyam) stays fit (swasthya); one who ignores this falls ill even with the best facilities.

Key points to remember

```
Human being = "I" (sentient) + Body (material)
"I" needs sukh (continuous, qualitative);  Body needs suvidha (temporary, quantitative)
"I" = seer, doer, enjoyer;  Body = instrument
Sanyam (in I) → Swasthya (in Body)
Physical needs are limited → prosperity = producing more than needed
```

⭐ Important questions

1. Explain the human being as the co-existence of the self ("I") and the Body. *Use the comparison table.*
2. Differentiate the needs of the self and the needs of the Body (sukh and suvidha).
3. Explain the Body as an instrument of "I"; describe the activities of "I" and harmony in "I".
4. What are Sanyam and Swasthya? Explain the programmes to ensure them.
5. Explain the meaning of prosperity and the correct appraisal of physical needs.

UNIT 3 Understanding Harmony in the Family and Society

✍️ 2-mark definitions

- **Family:** the basic unit of human interaction, where we first learn relationships.
- **Nyaya (justice):** recognising the relationship, fulfilling it with the right values, evaluating the fulfilment, and ensuring mutual happiness (**ubhay-tripti**).
- **Ubhay-tripti:** mutual fulfilment: both people in the relationship feel happy.
- **Vishwas (trust):** being assured that the other wants to make me happy and prosperous. It is the foundational value of relationship.
- **Intention vs competence:** intention is what one wants to do (always to make the other happy); competence is the ability to actually do it. **We judge ourselves by intention and others by competence**, which leads to mistrust.
- **Samman (respect):** right evaluation: seeing the other as similar to me (same aspirations, same "I"), not differentiating on the basis of body, wealth or beliefs.
- **Differentiation:** the opposite of respect: over-evaluation, under-evaluation or otherwise-evaluation of a person.
- **Akhand Samaj:** an undivided society, extending the feeling of relationship to every human.
- **Sarvabhaum Vyawastha:** universal order, extending harmony from family to the whole world family.

💡 Core concepts

Nine values in human relationships: 1. **Vishwas** (trust), 2. **Samman** (respect), 3. **Sneh** (affection), 4. **Mamta** (care), 5. **Vatsalya** (guidance), 6. **Shraddha** (reverence), 7. **Gaurav** (glory), 8. **Kritagyata** (gratitude), 9. **Prem** (love, the complete value).

Trust: intention vs competence

	Intention	Competence
Meaning	Wish to make the other happy	Ability to actually do it
Usually	Same in everyone	Differs from person to person
Mistake we make	We doubt others' intention when their competence fails	
Right approach	Trust the intention; help improve competence	

Respect vs differentiation: differentiation happens on the basis of **body** (gender, age, race, physical strength), **physical facilities** (wealth, post) and **beliefs** (isms, sects). Respect means recognizing that every human has the same "I" with the same basic aspiration.

Comprehensive human goals (harmony in society)

Goal	Meaning
Samadhan	Right understanding (resolution) in every individual

Goal	Meaning
Samridhi	Prosperity in every family
Abhay	Fearlessness (trust) in society
Sah-astitva	Co-existence in nature (existence)

Five dimensions of a human order: (1) education–right living (*shiksha–sanskar*), (2) health–self-regulation (*swasthya–sanyam*), (3) justice–preservation (*nyaya–suraksha*), (4) production–work (*utpadan–karya*), (5) exchange–storage (*vinimay–kosh*).

From family to world family: harmony in the family → harmony in society (family as the basic unit) → **Akhand Samaj** (undivided society) → **Sarvabhaum Vyawastha** (universal human order). Present problems (wars, terrorism, exploitation) come from fear and mistrust; the solution is right understanding and justice at every level.

Examples from life

- A friend forgets your birthday. You assume "they don't care" (doubting intention), when actually they were busy with exams (a competence issue). Trusting their intention preserves the relationship.
- Respecting a security guard and a manager equally, as human beings, is samman; treating them differently by post is differentiation.

Key points to remember

```
Family = basic unit;  Nyaya → ubhay-tripti (mutual happiness)
Trust = foundational value;  judge intention, help competence
Respect = right evaluation;  differentiation by body, wealth, belief
9 values: vishwas, samman, sneh, mamta, vatsalya, shraddha, gaurav, kritagyata, prem
Goals: samadhan (individual), samridhi (family), abhay (society), sah-astitva (nature)
Undivided society (akhand samaj) → universal order (sarvabhaum vyawastha)
```

⭐ Important questions

1. Explain harmony in the family as the basic unit of human interaction, and the meaning of justice (nyaya) and ubhay-tripti.
2. Explain trust as the foundational value; differentiate intention and competence with examples.
3. Explain respect as right evaluation; how does differentiation arise?
4. List and explain the values in human relationships.
5. Explain the comprehensive human goals (samadhan, samridhi, abhay, sah-astitva) and the dimensions of human order.
6. Explain undivided society and universal order (from family to world family).

UNIT 4 Understanding Harmony in Nature and Existence

✍️ 2-mark definitions

- **Nature:** the collection of all units (material, plant, animal and human) existing in space.
- **Existence:** nature submerged in all-pervasive space: **existence = co-existence (sah-astitva) of units in space**.
- **Four orders of nature: material order** (soil, water, metals), **plant/bio order** (grass, trees), **animal order** (birds, animals), **human order** (human beings).
- **Interconnectedness:** every unit is related to every other unit.
- **Mutual fulfilment:** each order fulfils the others (e.g. soil nourishes plants, plants give oxygen and food).
- **Recyclability:** materials in nature are reused in cycles (e.g. leaves decompose into soil).
- **Self-regulation:** each order maintains its own balance (e.g. population of species balances naturally).
- **Space:** all-pervasive, unbounded, transparent, energy-in-equilibrium, in which all units exist.

💡 Core concepts

The four orders and their mutual fulfilment

Order	Fulfils others by	Is it mutually fulfilling to all?
Material	Provides soil, water, minerals to plants, animals, humans	Yes
Plant	Gives food, oxygen, medicine; returns nutrients to soil	Yes
Animal	Helps pollination, carries seeds, manure; companionship to humans	Yes
Human	Should nurture and protect the other three orders	**Not yet**: humans often exploit, pollute and destroy

The human order is the only one that has not yet fulfilled its role; doing so needs **right understanding**.

Recyclability and self-regulation: nature works in cycles (water, carbon, nitrogen cycles) with no waste, and each order regulates itself. Human activity that ignores this (non-biodegradable waste, over-extraction) breaks the balance and causes pollution, resource depletion and climate change.

Existence as co-existence: units are **submerged in space**; they are energized, self-organized and recognise and fulfil their relationship with other units. Seeing existence as co-existence leads to living in harmony rather than domination.

Holistic perception of harmony: harmony at the four levels (self → family → society → nature) is connected; understanding one helps the others. Holistic understanding is the basis for a sustainable way of life.

Examples from life

- A tree uses soil and water (material order), gives fruit to birds (animal order) and humans, and its fallen leaves become compost: interconnectedness and recyclability.
- Plastic waste does not return to nature's cycles and harms animals and soil: a human failure of mutual fulfilment.

Key points to remember

```
Four orders: material, plant, animal, human
Nature has: interconnectedness, mutual fulfilment, recyclability, self-regulation
Existence = co-existence (sah-astitva) of units in space
Only the human order is not yet mutually fulfilling → needs right understanding
```

⭐ Important questions

1. Explain the four orders of nature and the interconnectedness and mutual fulfilment among them. *Table + diagram of four overlapping orders.*
2. Explain recyclability and self-regulation in nature, with examples.
3. Explain existence as co-existence of units in space.
4. Why is the human order not yet in harmony with the other orders? What is required?
5. Explain the holistic perception of harmony at all levels of existence.

NEXT ENGINEER _sheerat_

UNIT 5 Holistic Understanding and Professional Ethics

✍️ 2-mark definitions

- **Ethical human conduct:** behaviour and work based on right understanding, naturally acceptable and leading to mutual fulfilment. It includes **values, policy and character**.
- **Definitiveness of ethical conduct:** right understanding makes conduct definite (the same in all situations), instead of changing with mood, fear or greed.
- **Humanistic education:** education that develops right understanding along with skills, aiming at universal human order.
- **Humanistic constitution:** a constitution based on human values, ensuring the five dimensions of human order for all.
- **Professional ethics:** applying values and ethical conduct in one's profession; competence to use professional skills for universal human order.
- **Holistic technology:** technology that is people-friendly, eco-friendly, and supports mutual fulfilment and recyclability (e.g. renewable energy, organic farming).
- **Universal human order:** a world in which all humans live with right understanding, prosperity, fearlessness and co-existence with nature.

💡 Core concepts

Natural acceptance of human values: every human naturally accepts relationship, trust, respect and co-existence; this is the basis for universally accepted ethics, not imposed rules.

Competence in professional ethics

1. Ability to use professional competence to **augment universal human order** (not just profit).
2. Ability to identify the scope and characteristics of **people-friendly and eco-friendly** production systems.
3. Ability to identify and develop **appropriate technologies and management patterns**.

Holistic vs present-day systems

Present systems	Holistic systems
Profit and competition first	Mutual fulfilment and cooperation
Linear: take → make → dispose	Cyclic: renewable, recyclable
Centralized, large-scale	Decentralized, appropriate scale
Exploit nature and labour	Nurture nature and people
Examples: fossil-fuel plants, intensive chemical farming	Examples: solar/wind, natural farming, cooperatives, Amul model

Case studies of holistic models (quote one or two): organic/natural farming (Subhash Palekar's zero-budget natural farming), cooperative dairy (Amul), rainwater harvesting and watershed projects,

renewable energy villages, appropriate technology for rural industries.

Unethical practices in professions (examples): cutting corners on safety and quality, bribery and corruption, plagiarism and IP theft, misuse of data, environmental violations, conflicts of interest. Root cause: lack of right understanding and treating profit as the only goal.

Strategy for transition to universal human order

- **Individual level:** become socially and ecologically responsible engineers, technologists and managers; practise self-exploration; choose ethical work.
- **Society level:** build mutually enriching institutions and organizations; spread humanistic education; promote holistic technologies and policies.

Examples from life

- An engineer refusing to sign off on a bridge that does not meet safety standards, despite pressure, shows definiteness of ethical conduct.
- A software company that uses customer data only with consent and builds accessible products is applying professional ethics as people-friendly technology.

Key points to remember

```
Ethical conduct = values + policy + character;  definite with right understanding
Humanistic education + humanistic constitution → universal human order
Professional competence: augment human order; people-friendly, eco-friendly systems
Holistic tech: cyclic, renewable, decentralized, cooperative
Transition: individual (responsible professionals) + society (mutually enriching institutions)
```

⭐ Important questions

1. Explain the natural acceptance of human values and the definitiveness of ethical human conduct.
2. Explain the basis for humanistic education, humanistic constitution and universal human order.
3. What is competence in professional ethics? Explain with examples.
4. Explain people-friendly and eco-friendly technologies and management models, with case studies.
5. Explain the strategy for transition from the present state to universal human order at individual and society level.

Last-hour checklist

Read only this in the last hour before the paper.

- ☐ Value education: need, guidelines, content, process.
- ☐ Self-exploration: natural acceptance + experiential validation.
- ☐ Basic aspiration and three requirements with priority.
- ☐ Present scenario: only physical facilities → animal consciousness; right priority → human consciousness.

- ☐ "I" vs Body table; sukh vs suvidha; seer–doer–enjoyer.
- ☐ Sanyam → Swasthya; programmes.
- ☐ Nyaya, ubhay-tripti; trust: intention vs competence; respect vs differentiation.
- ☐ Nine values in relationships.
- ☐ Samadhan, samridhi, abhay, sah-astitva; five dimensions of human order.
- ☐ Four orders of nature; mutual fulfilment, recyclability, self-regulation.
- ☐ Ethical conduct; holistic technologies; transition strategy.

Books prescribed by AKTU

These are the books named in the AKTU syllabus. Buy or borrow them through a bookshop or your college library. The WorldCat links show which libraries hold a copy.

#	Title	Author(s)	Publisher	Where to get it legally
1	A Foundation Course in Human Values and Professional Ethics (text book, 2nd rev. ed.)	R. R. Gaur, R. Asthana, G. P. Bagaria	Excel Books	WorldCat
2	Energy and Equity	Ivan Illich	Harper & Row	WorldCat
3	Small Is Beautiful	E. F. Schumacher	Blond & Briggs	WorldCat
4	How the Other Half Dies	Susan George	Penguin	WorldCat
5	The Limits to Growth	D. H. Meadows, D. L. Meadows, J. Randers, W. W. Behrens III	Universe Books	WorldCat
6	Jeevan Vidya: Ek Parichay	A. Nagraj	Divya Path Sansthan	WorldCat
7	Science and Humanism	P. L. Dhar, R. R. Gaur	Commonwealth Publishers	WorldCat
8	Human Values	A. N. Tripathi	New Age International	WorldCat
9	Fundamentals of Ethics for Scientists and Engineers	E. G. Seebauer, R. L. Barry	Oxford University Press	WorldCat
10	Engineering Ethics (including Human Values)	M. Govindarajan, S. Natarajan, V. S. Senthil Kumar	PHI Learning	WorldCat
11	Foundations of Ethics and Management	B. P. Banerjee	Excel Books	WorldCat
12	Indian Ethos and Modern Management	B. L. Bajpai	New Royal Book Co.	WorldCat

AICTE publishes free UHV teaching material on its official website; check there for the latest editions.

Official syllabus: [AKTU B.Tech 2nd year CSE syllabus, 2023-24 scheme (PDF)]

NEXT ENGINEER
sheerat